Smart Agriculture AI — Comprehensive Farm Report

Generated: February 27, 2026 at 15:40

1. Soil Analysis

Parameter	Value
Detected Soil Type	Clay
Detection Confidence	99.7%
Detection Method	EfficientNetV2 + SVM

2. Current Soil Conditions (IoT Sensor)

Parameter	Value	Status
Soil Temperature	24.0°C	Normal
Soil Humidity	40.0%	Normal
Soil Moisture	88%	Healthy
Irrigation Status	NO ACTION	

3. Weather Conditions

Parameter	Value
City	Bhopal, India
Outside Temperature	30.6°C
Humidity	64%
Rainfall	0.0 mm/hr
Wind Speed	10.0 km/h

4. Field Details

Total Area: **10,000 m²** (2.47 acres)

5. AI Crop Recommendation

Mode: **Multi-Crop (Zone Allocation)**

Crop	Area (ac)	Alloc %	Yield (kg)	Price/kg	Profit (■)	Rating

Rice	0.74	30.0%	1,350	■25	■33,750	Excellent
Sugarcane	0.74	30.0%	27,000	■3.5	■94,500	Excellent
Wheat	0.49	20.0%	750	■22	■16,500	Good
Beans	0.49	20.0%	600	■55	■33,000	Good
TOTAL	**2.47**	**100%**	**29,700**		**■177,750**	

6. Crop Selection Reasoning

Rice (Score: 0.9):
• [OK] Compatible with Clay soil
• [OK] Temperature 24.0°C ideal (range: 20-35°C)
• [OK] Soil moisture 88% ideal (range: 60-100%)
• [i] Current rainfall: 0.0mm/hr

Sugarcane (Score: 0.9):
• [OK] Compatible with Clay soil
• [OK] Temperature 24.0°C ideal (range: 20-30°C)
• [OK] Soil moisture 88% ideal (range: 60-90%)
• [i] Current rainfall: 0.0mm/hr

Wheat (Score: 0.6):
• [OK] Compatible with Clay soil
• [OK] Temperature 24.0°C ideal (range: 15-25°C)
• [!] Soil moisture 88% outside ideal (40-70%)
• [i] Current rainfall: 0.0mm/hr

Beans (Score: 0.6):
• [OK] Compatible with Clay soil
• [OK] Temperature 24.0°C ideal (range: 18-27°C)
• [!] Soil moisture 88% outside ideal (40-70%)
• [i] Current rainfall: 0.0mm/hr

Report generated by Smart Agriculture AI — IoT + Deep Learning System
Cost savings: Farmers typically spend ■10,000-20,000 on lab soil testing.